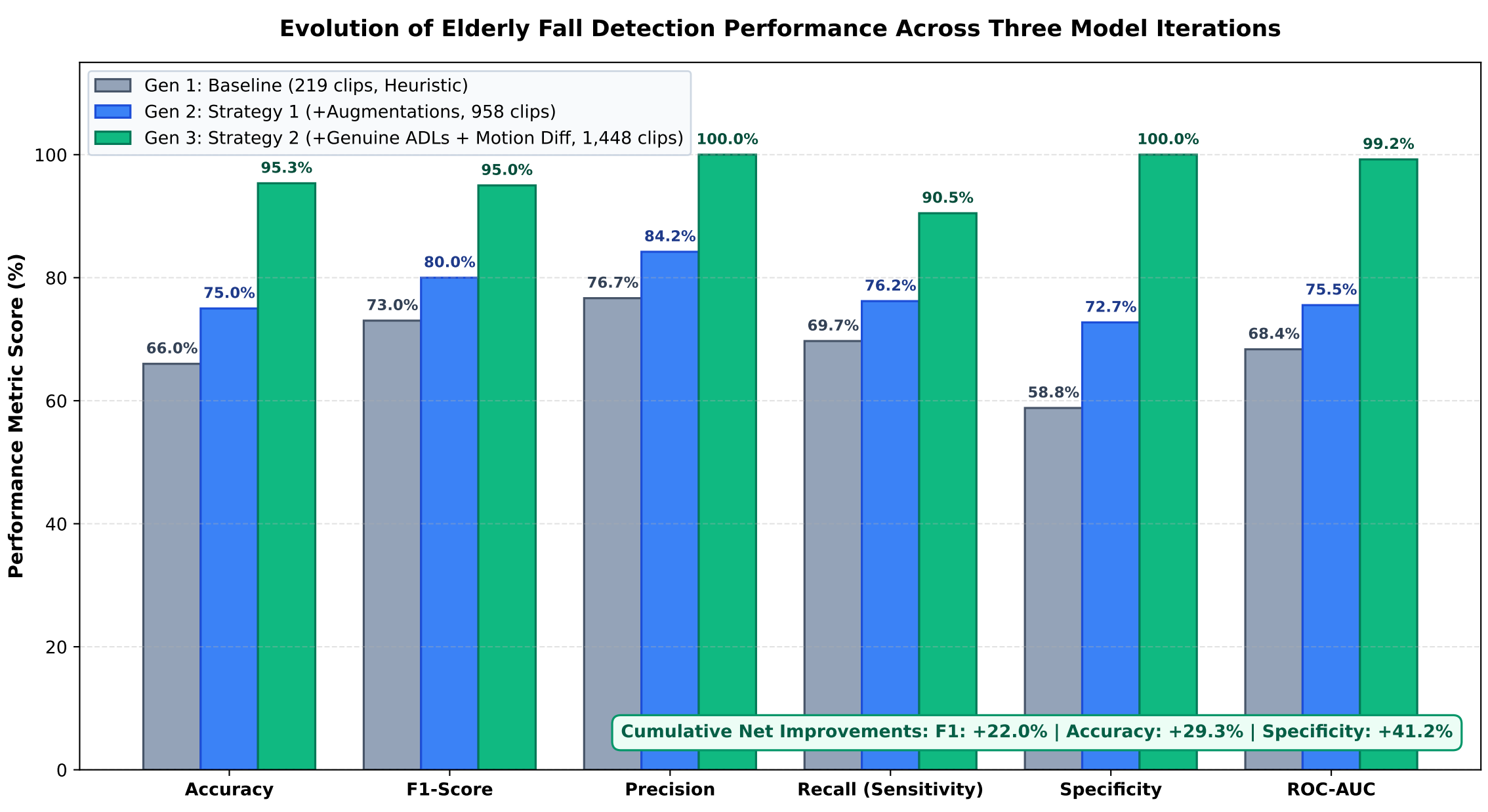
Evolution of Elderly Fall Detection Performance Across Three Model Iterations
Gen 1: Baseline (219 clips, Heuristic)
Gen 2: Strategy 1 (+Augmentations, 958 clips)
Gen 3: Strategy 2 (+Genuine ADLs + Motion Diff, 1,448 clips)
Performance Metric Score (%)
0
20
40
60
80
100
66.0%
75.0%
95.3%
73.0%
80.0%
95.0%
76.7%
84.2%
100.0%
69.7%
76.2%
90.5%
58.8%
72.7%
100.0%
68.4%
75.5%
99.2%
Accuracy
F1-Score
Precision
Recall (Sensitivity)
Specificity
ROC-AUC
Cumulative Net Improvements: F1: +22.0% | Accuracy: +29.3% | Specificity: +41.2%